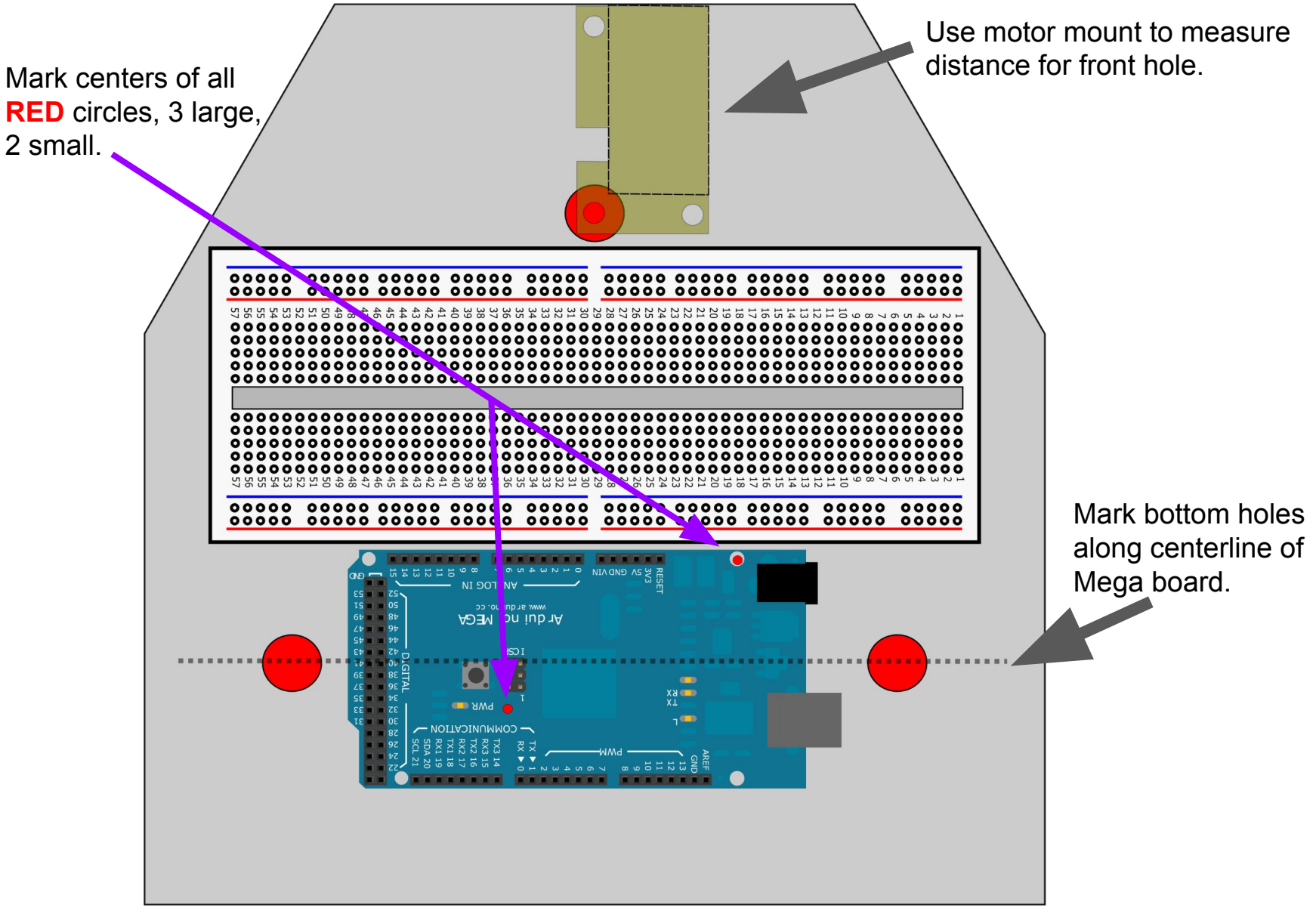
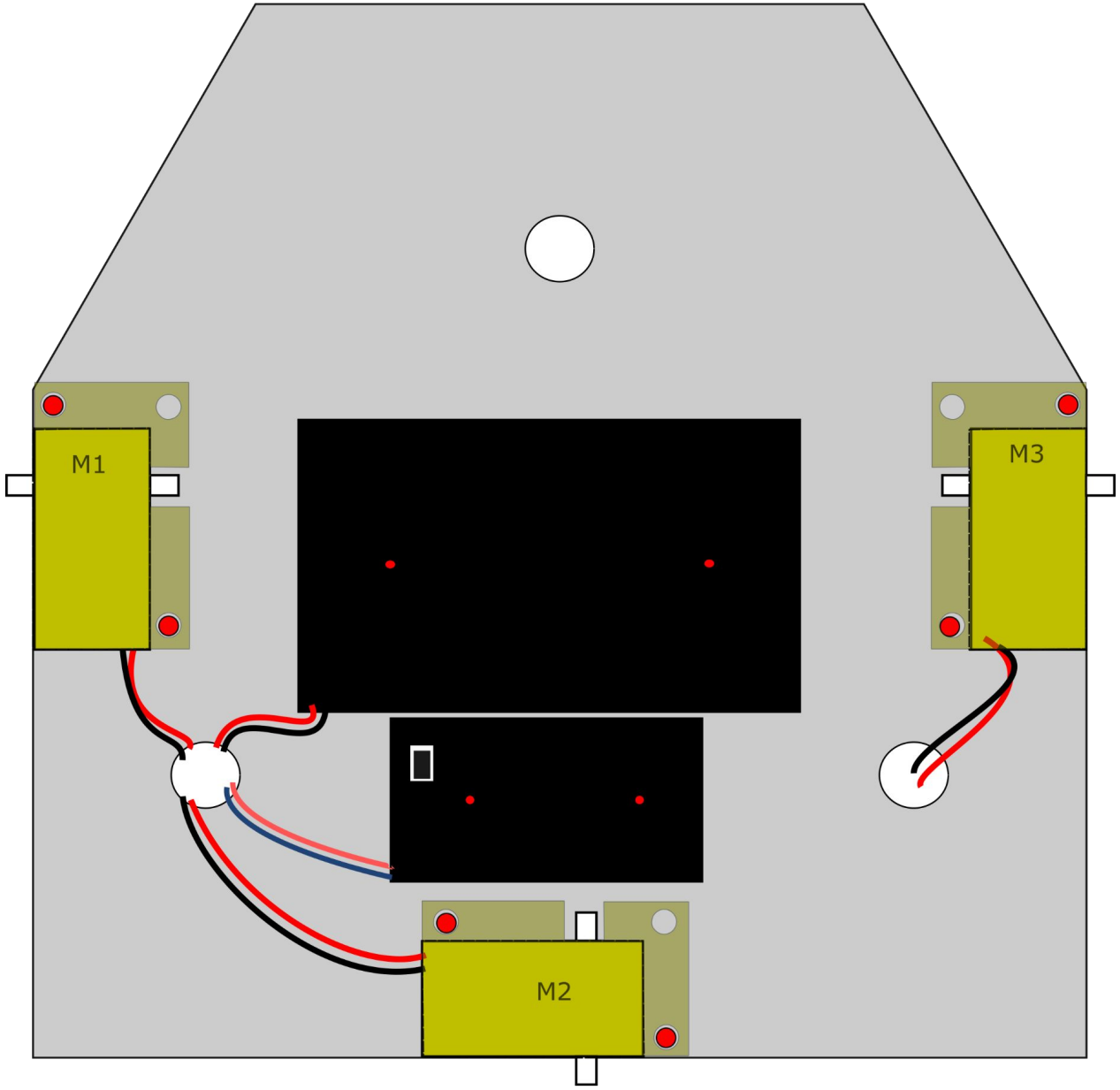
# BreadBoardBot

Layout and wiring,  rev. 2024b
8/7/2023

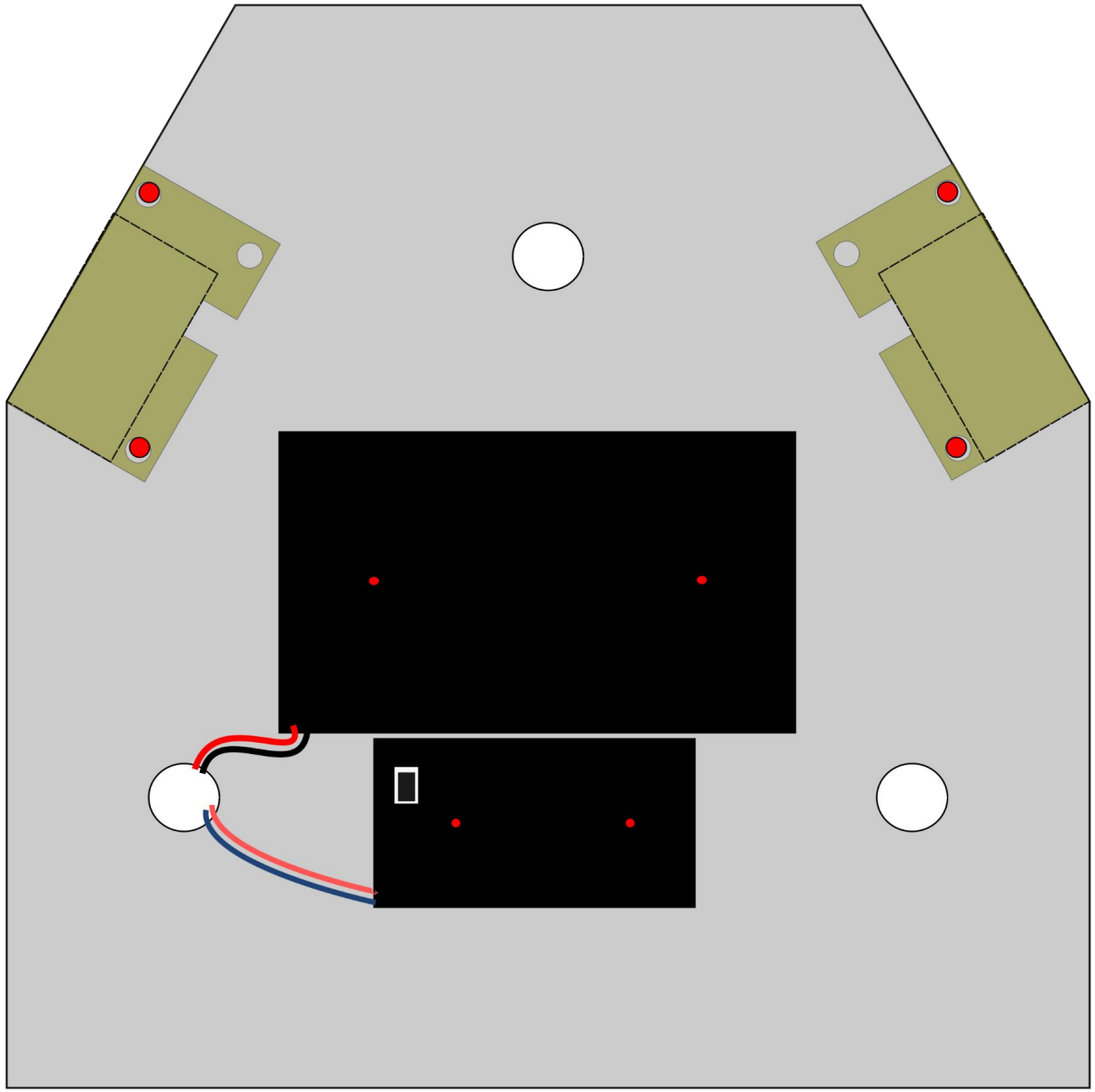
# B3 Top Layout: 3 large holes, 2 medium holes



# B3 Bottom Layout 1: 6 medium holes, 4 small holes



# B3 Bottom Layout 2: 4 more medium holes



## Drill holes, then…

- **Always wear safety glasses when using the drills.** This is non-negotiable.
- Use medium sized bit for all motor mount holes (10 of those) and, if desired, to drill pilot holes for the ½" big holes (3 of those).
- Drill six small holes, 2 marked on front for the Mega, and 4 on the back for both battery holders.
- Drill the three big holes, which is much easier with pilot holes.

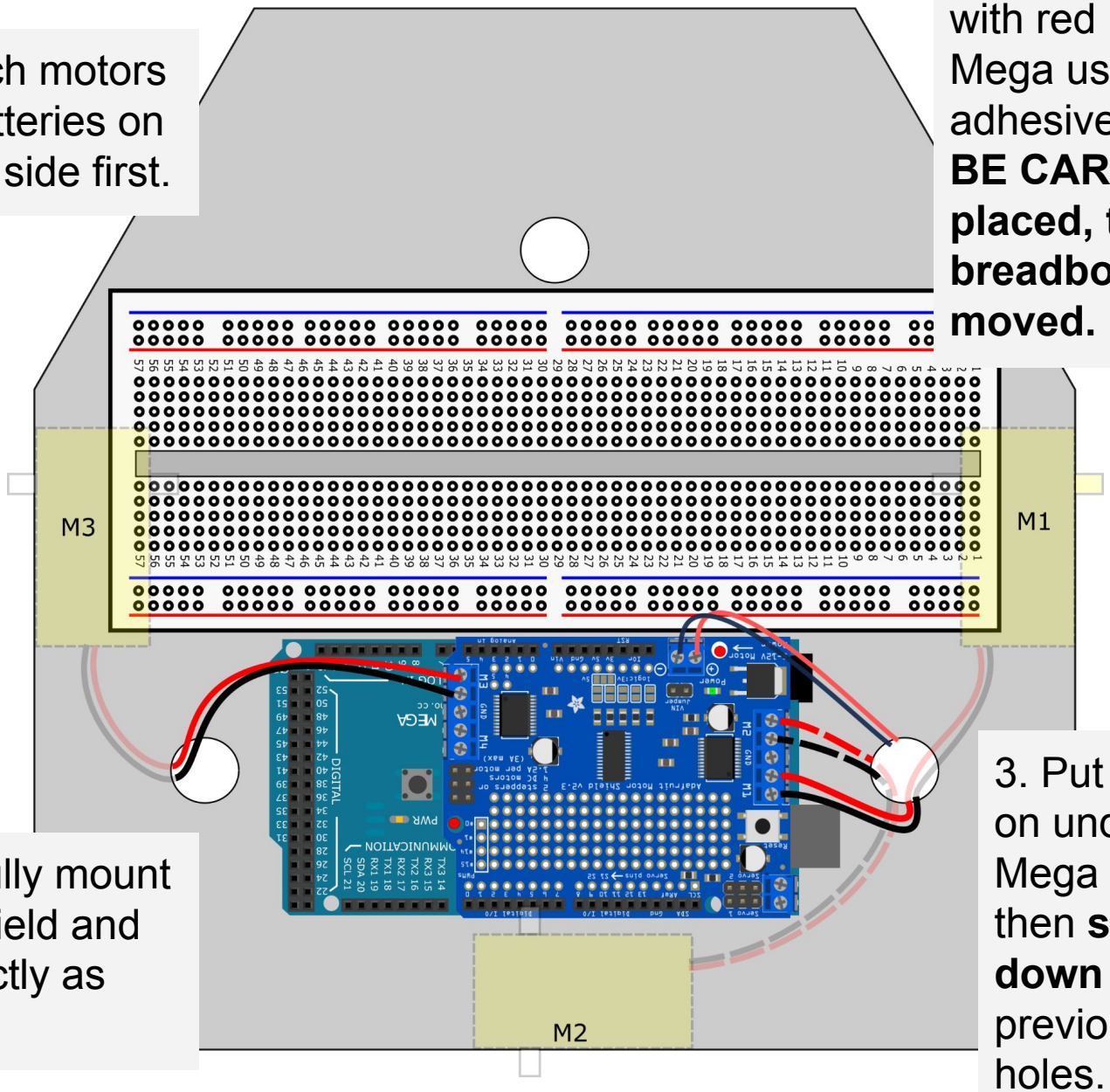
## Omniwheel assembly (demo)

- Place motors into motor mounts. Make sure the stub on side of motor fits into hole on side of motor mounts.
- Using the small white tubes and the long small screws (3 of each, one set per motor), place the tube **fully** on the outside motor axle (the side away from the rim of the motor mount.)
- Drill small hole through axle and tube and secure tube in place with screw.
- Get omniwheels (3) and the largest screws. Put a small amount of hot glue on the tube, push omni wheel over tube, and screw large screw into tube to expand the tube and provide grip for the wheel.

# B3 Begin Assembly (shield demo)

1. Attach motors and batteries on bottom side first.

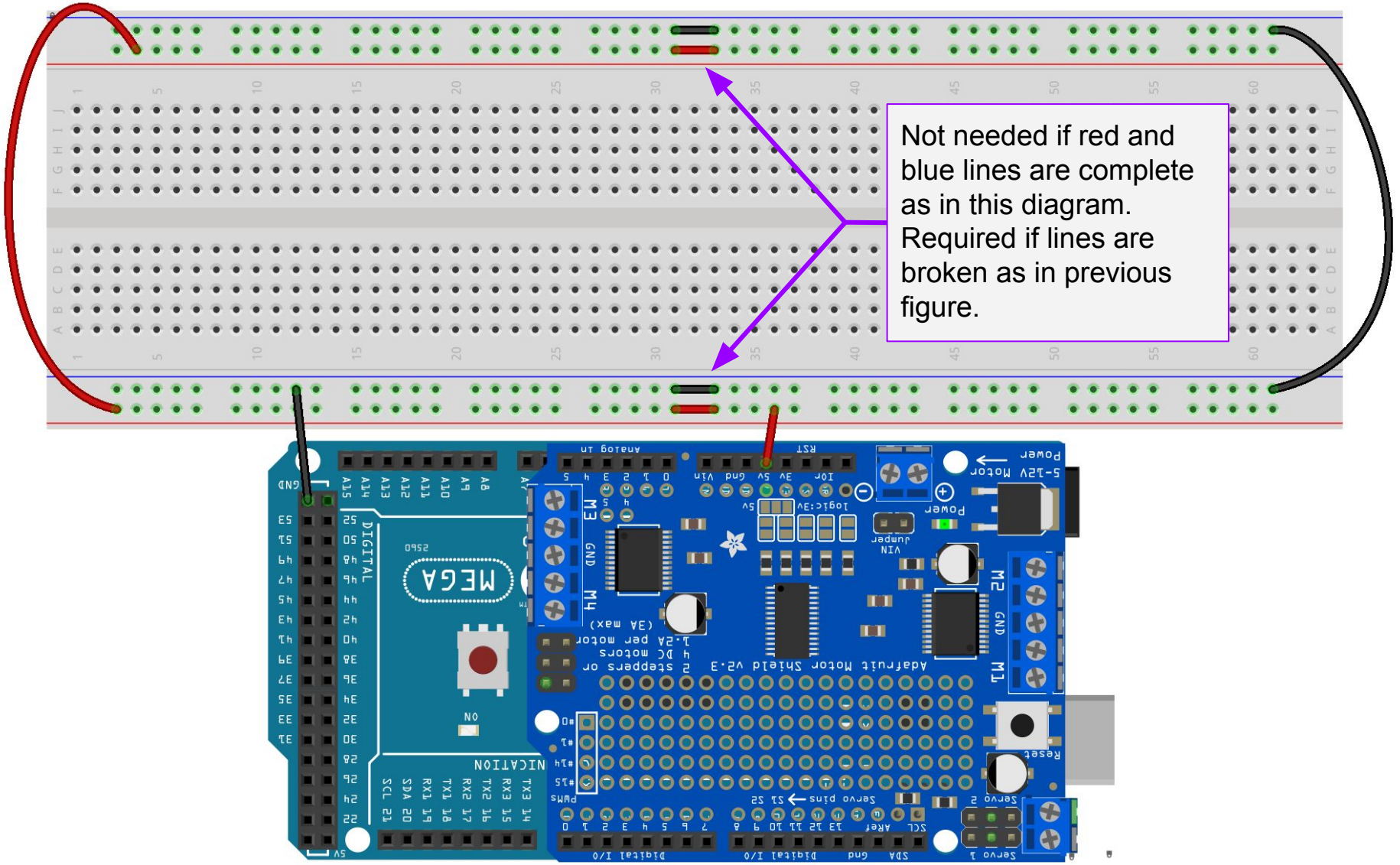
2. Attach breadboard with red line next to Mega using the adhesive on its back. **BE CAREFUL! Once placed, the breadboard can not be moved.**



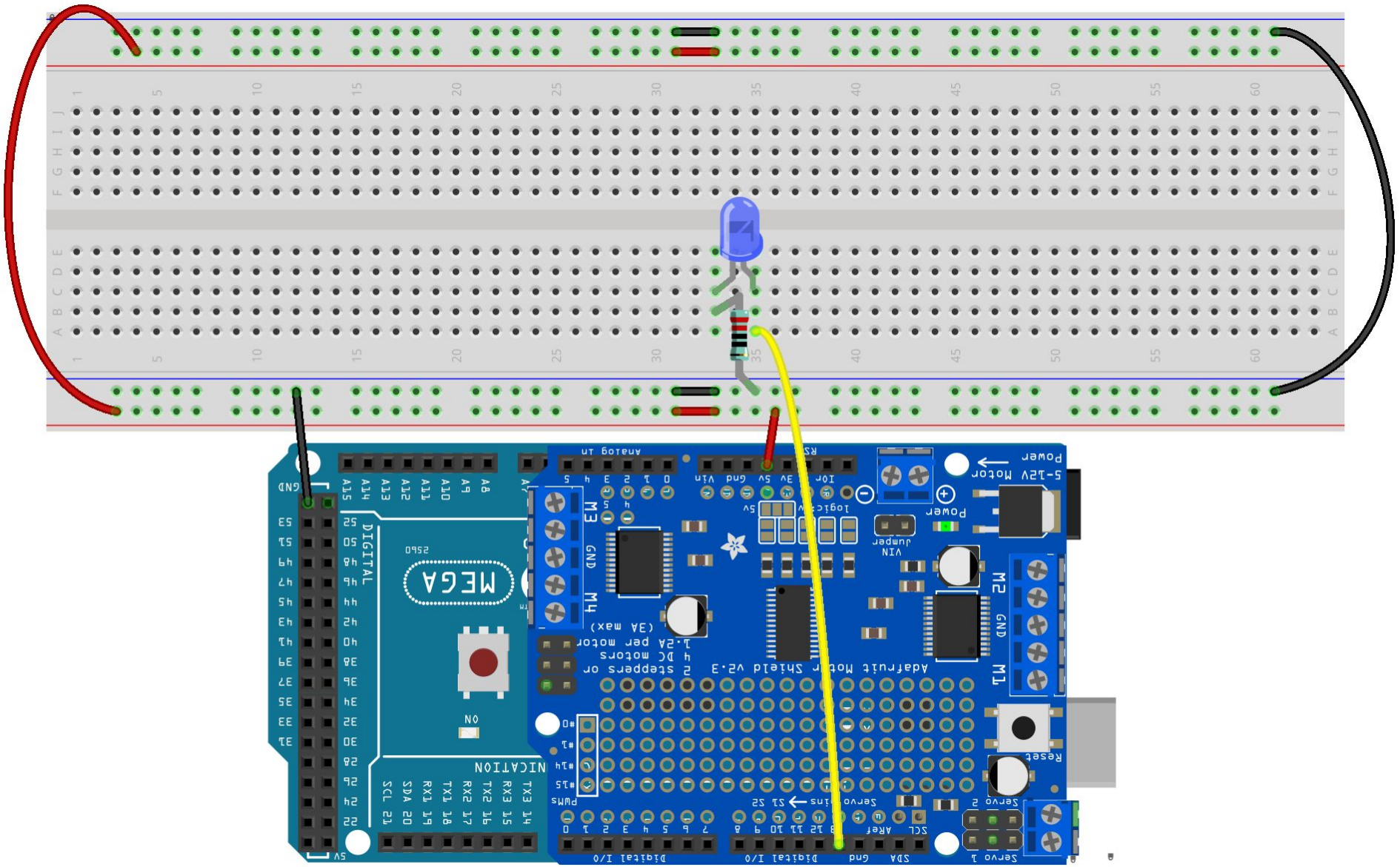

3. Put 3 balls of putty on underside of Mega as cushion, then **screw Mega down** through previously drilled holes.

4. Carefully mount motor shield and wire exactly as shown.

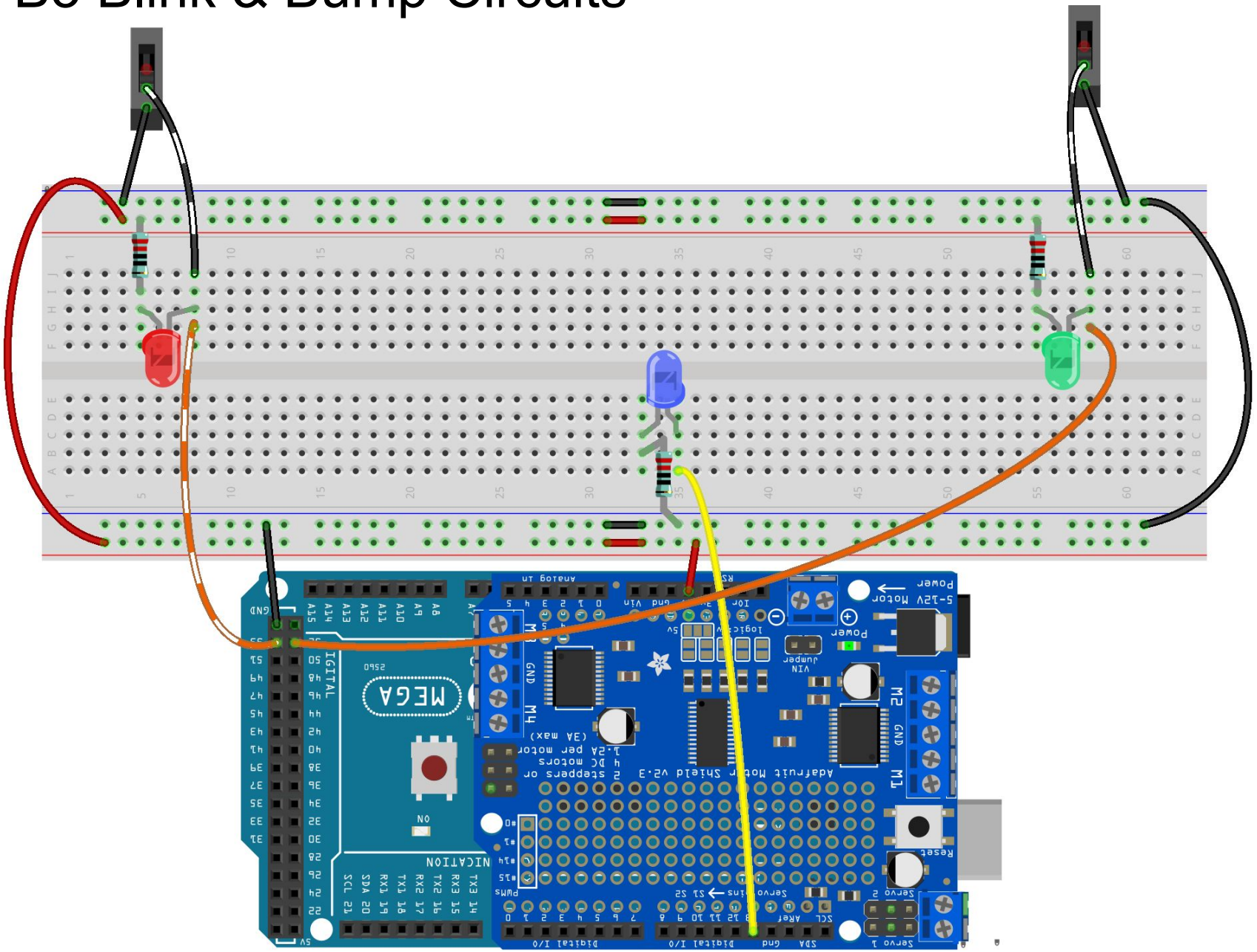
# B3 Power and Ground Circuits



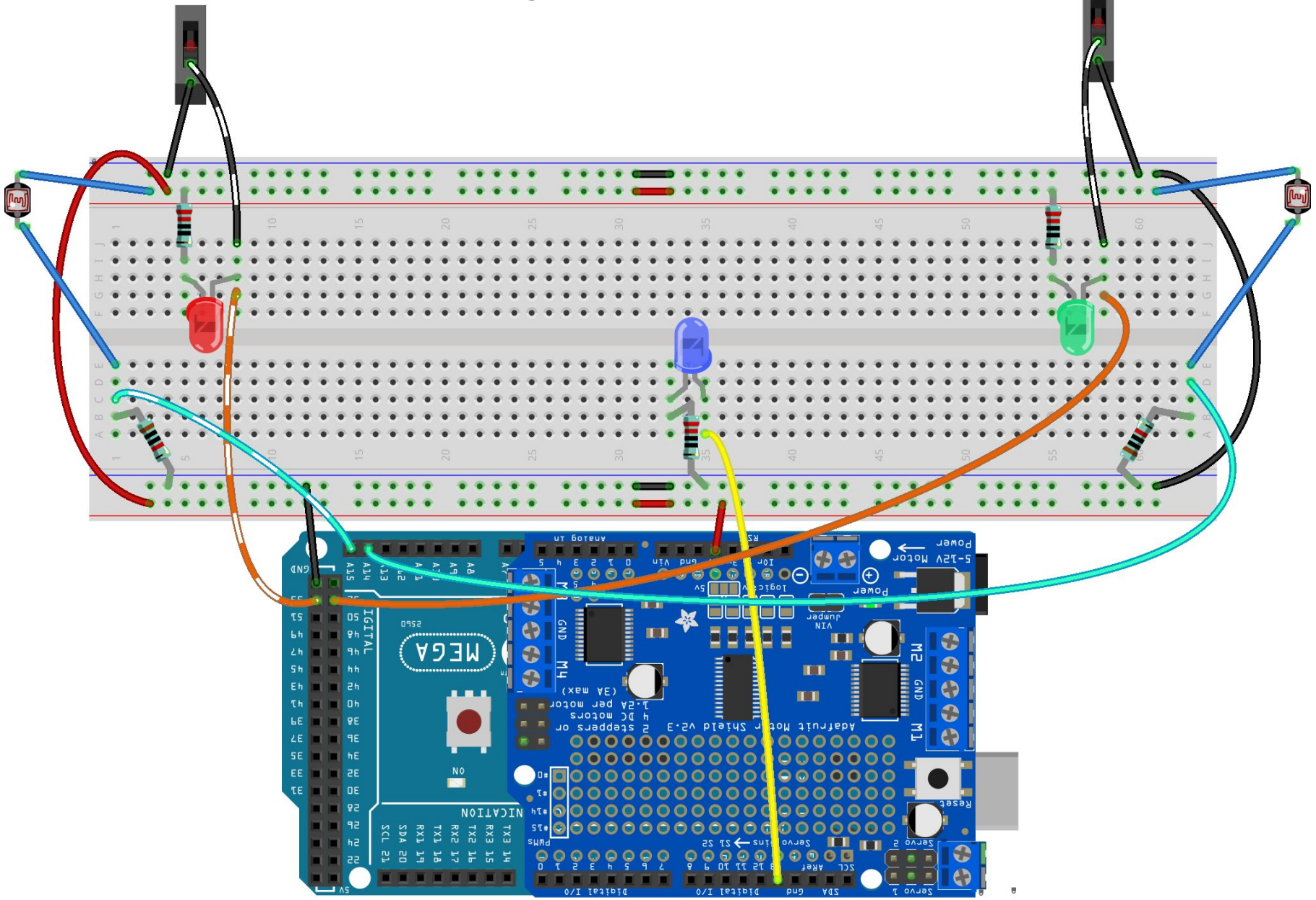
# B3 Blink Circuits



# B3 Blink & Bump Circuits

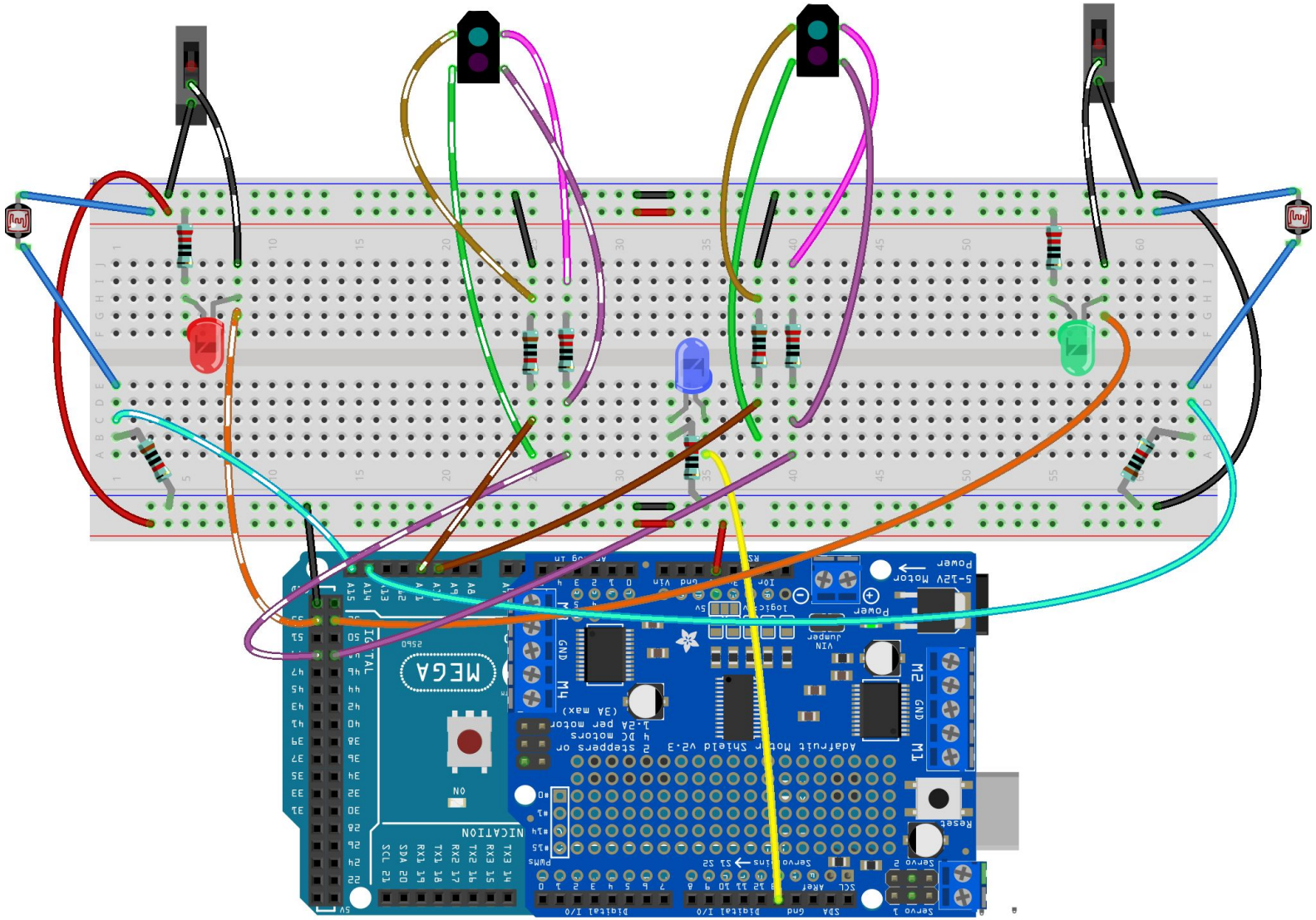


# B3 Blink, Bump & Light Detection Circuits



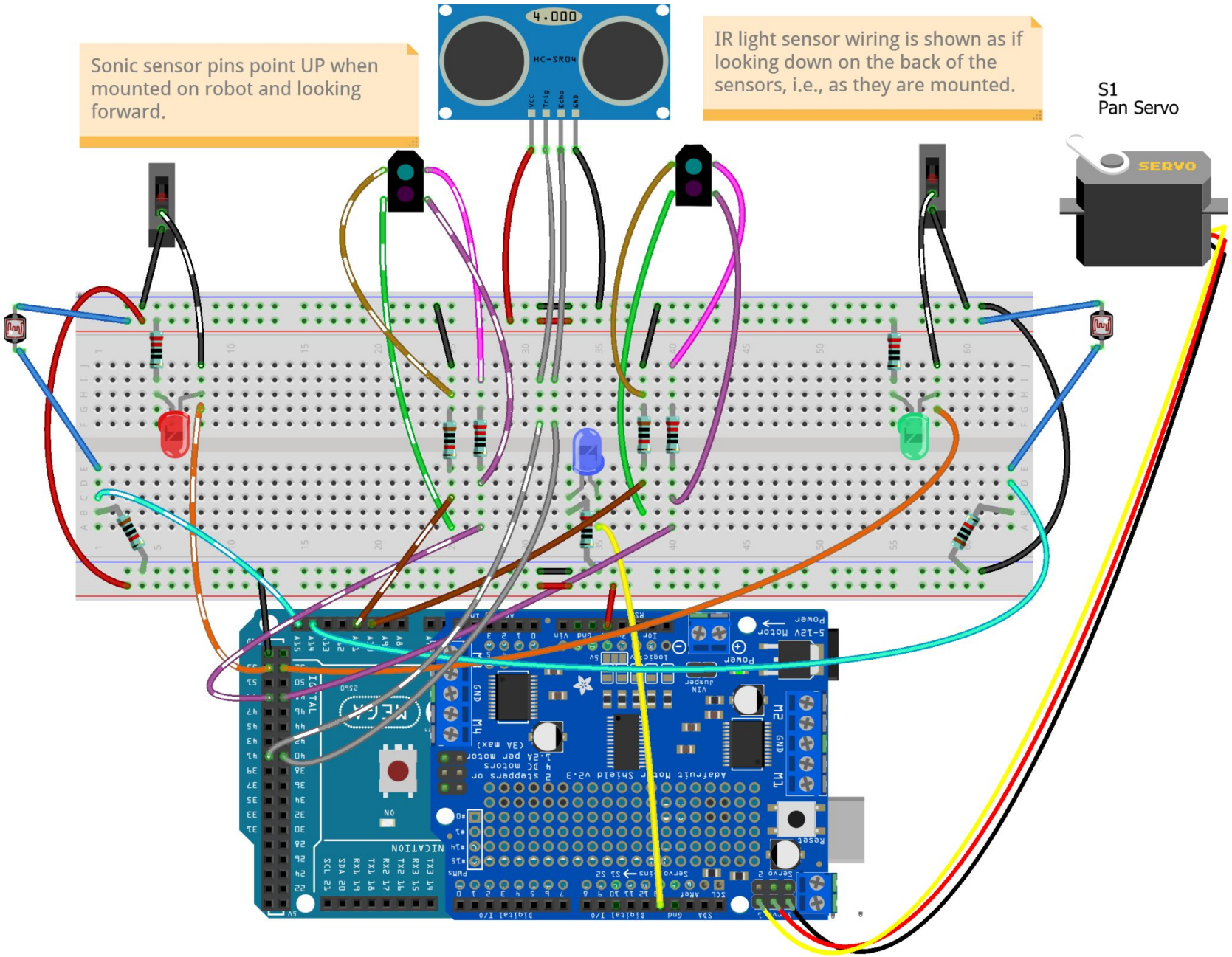
# B3 Blink, Bump, Light & IR Circuits

IR light sensor wiring is shown as if looking down on the back of the sensors, i.e., as they are mounted.



fritzing

# B3 Blink, Bump, Light, IR, Sonic & Servo Circuits



# B3 Bonus Circuits

